

- First I downloaded the binwalk git repository and unzipped the master. Using the commands **wget https://github.com/devttys0/binwalk/archive/master.zip** and **unzip master.zip** in order to analyse the firmware files and explore their contents.
- Then I installed binwalk using the command **sudo python3 setup.py install**
- Then I installed squashfs-tools using command **sudo apt-get install squashfs-tools** which allowed me to extract the squashfs file system.
- Then I used **binwalk-Me<filename>** command on the Firmware1.bin file to extract the contents of the firmware binaries. This is a standard and effective approach to identify and extract embedded files within firmware images.

```
osboxes@osboxes:~/Desktop/ESS$ ls
Binwalk  Firmware1.bin  Firmware2.bin  Firmware3.bin
osboxes@osboxes:~/Desktop/ESS$ binwalk -Me Firmware1.bin

Scan Time:      2023-12-01 10:54:28
Target File:    /home/osboxes/Desktop/ESS/Firmware1.bin
MD5 Checksum:   d393478d1c9b0853c3f37124148ef864
Signatures:     411

DECIMAL          HEXADECIMAL      DESCRIPTION
-----
11288            0x2C18           LZMA compressed data, properties: 0x5D, dictionary
size: 8388608 bytes, uncompressed size: 3830348 bytes

WARNING: Extractor.execute failed to run external extractor 'sasquatch -p 1 -le
-d 'squashfs-root-0' '%e': [Errno 2] No such file or directory: 'sasquatch', 's
asquatch -p 1 -le -d 'squashfs-root-0' '%e' might not be installed correctly

WARNING: Extractor.execute failed to run external extractor 'sasquatch -p 1 -be
-d 'squashfs-root-0' '%e': [Errno 2] No such file or directory: 'sasquatch', 's
asquatch -p 1 -be -d 'squashfs-root-0' '%e' might not be installed correctly

WARNING: Symlink points outside of the extraction directory: /home/osboxes/Deskt
op/ESS/_Firmware1.bin.extracted/squashfs-root/mnt -> /tmp/mnt; changing link tar
get to /dev/null for security purposes.

WARNING: Symlink points outside of the extraction directory: /home/osboxes/Deskt
op/ESS/_Firmware1.bin.extracted/squashfs-root/var -> /tmp/var; changing link tar
get to /dev/null for security purposes.

WARNING: Symlink points outside of the extraction directory: /home/osboxes/Deskt
op/ESS/_Firmware1.bin.extracted/squashfs-root/lib/modules/build -> /home/builder
/ramdisk/fwbuild-DIR_615S-bceef5e769d54187a8ede65abf6f1862/output/build/kernel;
changing link target to /dev/null for security purposes.

WARNING: Symlink points outside of the extraction directory: /home/osboxes/Deskt
op/ESS/_Firmware1.bin.extracted/squashfs-root/lib/modules/source -> /home/buide
r/ramdisk/fwbuild-DIR_615S-bceef5e769d54187a8ede65abf6f1862/output/build/kernel;
changing link target to /dev/null for security purposes.

WARNING: Symlink points outside of the extraction directory: /home/osboxes/Deskt
op/ESS/_Firmware1.bin.extracted/squashfs-root/usr/bin/iptables-xml -> /usr/sbin/
xtables-multi; changing link target to /dev/null for security purposes.
```

```
WARNING: Symlink points outside of the extraction directory: /home/osboxes/Desktop/ESS/_Firmware1.bin.extracted/squashfs-root/etc/hosts -> /tmp/hosts; changing link target to /dev/null for security purposes.
1179648      0x120000      Squashfs filesystem, little endian, version 4.0, c
ompression:lzma, size: 2642360 bytes, 1475 inodes, blocksize: 131072 bytes, crea
ted: 2017-12-14 10:41:00
```

```
Scan Time:      2023-12-01 10:54:30
Target File:    /home/osboxes/Desktop/ESS/_Firmware1.bin.extracted/2C18
MD5 Checksum:  feabdbc908516ca4f7a4d3f05d5f9f8
Signatures:    411
```

DECIMAL	HEXADECIMAL	DESCRIPTION
3100704	0x2F5020	Linux kernel version 2.6.30
3132240	0x2FCB50	SHA256 hash constants, big endian
3260104	0x31BEC8	Unix path: /etc/Wireless/RTL8192CD.dat
3260254	0x31BF5E	Unix path: /etc/Wireless/RTL8192CD.dat
3260338	0x31BFB2	Unix path: /etc/Wireless/RTL8192CD.dat Success
3310200	0x328278	Neighborly text, "NeighborSolicitsip6_tables: (C
2000-2006 Netfilter Core Team"		
3310220	0x32828C	Neighborly text, "NeighborAdvertisements-2006 Netf
ilter Core Team"		
3312619	0x328BEB	Neighborly text, "neighbor %.2x%.2x%.2x%.2x%.2x
:%.2x%.2x%.2x%.2x lost on port %d(%s)(%s)"		
3319183	0x32A58F	HTML document header
3319346	0x32A632	HTML document footer
3470049	0x34F2E1	Intel x86 or x64 microcode, sig 0x0205080d, pf_mas
k 0x14191e21, 1B1E-12-17, rev 0x1f000000, size 1		
3470065	0x34F2F1	Intel x86 or x64 microcode, sig 0x0305090e, pf_mas
k 0x161c2225, 1E21-14-19, rev 0x22000000, size 1		
3475888	0x3509B0	AES S-Box

- Then I Git cloned the firmwalker repository using the command **git clone** <https://github.com/craigz28/firmwalker.git> to explore the extracted firmware directory and search for potential hints or files of interest.

```
osboxes@osboxes:~/Desktop/ESS/_Firmware1.bin.extracted$ git clone https://github
.com/craigz28/firmwalker.git
Cloning into 'firmwalker'...
remote: Enumerating objects: 316, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 316 (delta 0), reused 3 (delta 0), pack-reused 312
Receiving objects: 100% (316/316), 145.51 KiB | 1.39 MiB/s, done.
Resolving deltas: 100% (142/142), done.
```

- Then I used **firmware.sh <filename of extracted firmware>** command to look for files of interest.

```
osboxes@osboxes:~/Desktop/ESS/_Firmware1.bin.extracted/firmwalker$ ./firmwalker.
sh _Firmware1.bin.extracted
***Firmware Directory***
_Firmware1.bin.extracted
***Search for password files***
##### passwd
d/squashfs-root/usr/bin/passwd
d/squashfs-root/etc/passwd

##### shadow

##### *.psk

***Search for Unix-MD5 hashes***

***Search for SSL related files***
##### *.crt

##### *.pem
d/squashfs-root/etc/server.pem
d/squashfs-root/etc/intercept_server.pem
Shodan cli not found.
Shodan cli not found.

##### *.cer

##### *.p7b

##### *.p12

##### *.key
d/squashfs-root/etc/server.key
d/squashfs-root/etc/intercept_server.key

***Search for SSH related files***
##### authorized_keys

##### *authorized_keys*

##### host key
```

- Then I used the command **grep** to search for specific patterns or keywords within the content of files.
- In particular I used this command **grep -r -i -e "Keyword" -e "Keyword" /path/to/search/directory** and for this case I used **grep -r -i -e "PRIVATE KEY" -e "ENCRYPTED PRIVATE KEY" /home/osboxes/Desktop/ESS/firmwalker/_Firmware1.bin.extracted** Binary file

```
osboxes@osboxes:~/Desktop/ESS/_Firmware1.bin.extracted/firmwalker$ grep -r -i -e
"PRIVATE KEY" -e "ENCRYPTED PRIVATE KEY" /home/osboxes/Desktop/ESS/firmwalker/_
Firmware1.bin.extracted
Binary file /home/osboxes/Desktop/ESS/firmwalker/_Firmware1.bin.extracted/squash
fs-root/usr/sbin/anweb matches
Binary file /home/osboxes/Desktop/ESS/firmwalker/_Firmware1.bin.extracted/squash
fs-root/usr/lib/libaxtls.so.1.2 matches
/home/osboxes/Desktop/ESS/firmwalker/_Firmware1.bin.extracted/squashfs-root/etc/
server.key:-----BEGIN RSA PRIVATE KEY-----
/home/osboxes/Desktop/ESS/firmwalker/_Firmware1.bin.extracted/squashfs-root/etc/
server.key:-----END RSA PRIVATE KEY-----
/home/osboxes/Desktop/ESS/firmwalker/_Firmware1.bin.extracted/squashfs-root/etc/
intercept_server.key:-----BEGIN RSA PRIVATE KEY-----
/home/osboxes/Desktop/ESS/firmwalker/_Firmware1.bin.extracted/squashfs-root/etc/
intercept_server.key:-----END RSA PRIVATE KEY-----
```

- Then to extract and view the content of these keys, I used the **cat** command.

```
osboxes@osboxes:~/Desktop/ESS/_Firmware1.bin.extracted/firmwalker$ cat /home/osb
oxes/Desktop/ESS/firmwalker/_Firmware1.bin.extracted/squashfs-root/etc/server.ke
y
-----BEGIN RSA PRIVATE KEY-----
MIICXgIBAAKBgQC/oHDhaquGCMh+z6HvLPdjeY8o/WJBCmyFC40zDCVM9ajyajfy
oRLLFIniis7EQ5K0aLcb65ZzZAHrZE8DFe/Ma74k1ppAqp7mEeU/nxSypo6qJGYn
dDoUSaaCLN0tcKem1XRzo2p/zhpLuoCoFd5pqB9+Zy6J2+BMLXT7mHALMwIDAQAB
AoGBAIm0CsoB9IZaE3IFVRHh11j1hqnnA89HZVpqRap0R0277w9cwx2pIC8tcDSK
mTEyecO6fjMjJbeAG10DdT/7aJjYVGzG4yJKhnk70l90i3XtzSBx3YOCRS++6yD
c4p03wFaRgeL+xFKm517hsT+JkpmsAvrLE2pTptCb/19Ao7JAEa3bvEZ4E0mj32
rP7aB+S/gmvVVVg1vSfflEPzU5rzy98Q3ZGz1A7nJ7kVgP2XWbfZZ05XTpYfpR2j
Z+CiUT0lZwJBAN09lsSNk6a2tOYprz+jyAOQedy3ASwBecJpkUh4htaYf90h1e2k
khn9QefMoYYgvoG34Nei13ndOJZKu1iEYFUCQQDQIel5Msn6ooFC37IvSKQP00Mz
6in/VUdD+AiWiN0hodBF+vdu7VJDcjNiMif4el+i3eGMDdTEe66f2jbixN+vAkeA
nJ0xslgqkpS6AEi350wQlf1bVtQxvy4YPHiwwOWZRIZULsNchkq+piky0113ZlSd
p4Q8ZUnAUWM54KJsdT/WtQJAcnHi+w1ju5xvQMSZVN8EofEaeBGjiIqoUw5Ej38f
Eg6b308gyftdLdPcyt35EriIKhyRpt8GHcGpTRMGXltJvw==
-----END RSA PRIVATE KEY-----
```

Private key in Firmware1.bin - -----BEGIN RSA PRIVATE KEY-----

```
MIICXgIBAAKBgQC/oHDhaquGCMh+z6HvLPdjeY8o/WJBCmyFC40zDCVM9ajyajfy
oRLLFIniis7EQ5K0aLcb65ZzZAHrZE8DFe/Ma74k1ppAqp7mEeU/nxSypo6qJGYn
dDoUSaaCLN0tcKem1XRzo2p/zhpLuoCoFd5pqB9+Zy6J2+BMLXT7mHALMwIDAQAB
AoGBAIm0CsoB9IZaE3IFVRHh11j1hqnnA89HZVpqRap0R0277w9cwx2pIC8tcDSK
mTEyecO6fjMjJbeAG10DdT/7aJjYVGzG4yJKhnk70l90i3XtzSBx3YOCRS++6yD
c4p03wFaRgeL+xFKm517hsT+JkpmsAvrLE2pTptCb/19Ao7JAEa3bvEZ4E0mj32
rP7aB+S/gmvVVVg1vSfflEPzU5rzy98Q3ZGz1A7nJ7kVgP2XWbfZZ05XTpYfpR2j
Z+CiUT0lZwJBAN09lsSNk6a2tOYprz+jyAOQedy3ASwBecJpkUh4htaYf90h1e2k
khn9QefMoYYgvoG34Nei13ndOJZKu1iEYFUCQQDQIel5Msn6ooFC37IvSKQP00Mz
6in/VUdD+AiWiN0hodBF+vdu7VJDcjNiMif4el+i3eGMDdTEe66f2jbixN+vAkeA
nJ0xslgqkpS6AEi350wQlf1bVtQxvy4YPHiwwOWZRIZULsNchkq+piky0113ZlSd
```


p4Q8ZUnAUWM54KJsdt/WtQJAcnHi+w1ju5xvQMSZVN8EofEaeBGjilqoUw5Ej38f
Eg6b308gyftdLdPcyt35EriIKhyRPt8GHcGPtRMGXltJvw==

-----END RSA PRIVATE KEY-----

Firmware 2

For Firmwar2.bin I used the exact same commands and following is the private keys I located:

```
osboxes@osboxes:~/Desktop/ESS/Question1/firmwalker$ grep -r -i -e "PRIVATE KEY"
-e "ENCRYPTED PRIVATE KEY" ~/Desktop/ESS/Question1/_Firmware3.bin.extracted
grep: /home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root
/sbin/wpa_cli: binary file matches
grep: /home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root
/sbin/tr069: binary file matches
grep: /home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root
/sbin/wpa_supplicant: binary file matches
grep: /home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root
/usr/sbin/anweb: binary file matches
grep: /home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root
/usr/bin/racoon: binary file matches
grep: /home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root
/usr/lib/libssl.so.1.0.0: binary file matches
grep: /home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root
/usr/lib/libcrypto.so.1.0.0: binary file matches
grep: /home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root
/usr/lib/libcurl.so.5.3.0: binary file matches
/home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root/etc/s
erver.key:-----BEGIN RSA PRIVATE KEY-----
/home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root/etc/s
erver.key:-----END RSA PRIVATE KEY-----
/home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root/etc/t
r069_key.pem:-----BEGIN PRIVATE KEY-----
/home/osboxes/Desktop/ESS/Question1/_Firmware3.bin.extracted/squashfs-root/etc/t
r069_key.pem:-----END PRIVATE KEY-----
```

```

osboxes@osboxes:~/Desktop/ESS/Question1/firmwalker$ cat /home/osboxes/Desktop/ESS/Question1/Firmware3.bin.extracted/squashfs-root/etc/server.key
-----BEGIN RSA PRIVATE KEY-----
MIICXgIBAAKBgQC/oHDhaquGCMh+z6HvLPdjeY8o/WJBCmyFC40zDCVM9ajyajfy
oRLLFIniis7EQ5K0aLcb65ZzZAHrZE8DFe/Ma74klppAqp7mEeU/nxSypo6qJGYn
dDoUSaaCLN0tcKem1XRzo2p/zhpLuoCoFd5pqB9+Zy6J2+BMLXT7mHALMwIDAQAB
AoGBAIm0CsoB9IZaE3IFVRHh1lj1hqnnA89HZVpqRap0R0277w9cwx2pIC8tcDSK
mTEyec06fjmJjbeAG10DdT/7aJjYVGGzG4yJKhnk70l90i3XtzSBx3Y0CRS++6yD
c4p03wFaRgeL+xFKm517hsT+JkpmsAvrLE2pTptCb/19Ao7JAE3bvEZ4E0mj32
rP7aB+S/gmvVVVglvSfflEPzU5rzy98Q3ZGz1A7nJ7kVgP2XWbfZZ05XTpYfpR2j
Z+CiUT0lZwJBAN09lsSNk6a2t0Yprz+jyA0Qedy3ASwBecJpkUh4htaYf90h1e2k
khn9QefMoYYgvoG34Nei13nd0JZKu1iEYFUCQQDQIel5Msn6ooFC37IvSKQP00Mz
6in/VUdD+AiWiN0hodBF+vdu7VJDcjNiMif4el+i3eGMDdTEe66f2jbixN+vAkEA
nJ0xslgqkpS6AEi350wQlf1bVtQxvy4YPHiwwOWZRiZULsNchkg+piky0113ZlSd
p4Q8ZUNAUWM54KJsdt/WtQJAcnHi+w1ju5xvQMSZVN8EofEaeBGjiIqoUw5Ej38f
Eg6b308gyftdLdPcyt35EriIKhyRpt8GHcGPtRMGXltJvw==
-----END RSA PRIVATE KEY-----

```

Question 5

1) Explain what is meant by a Canary, and then evaluate the effectiveness and limitations of Canaries as a defence versus attacks on all types of stack buffer overflow vulnerabilities.

A Canary refers to a security mechanism that places a token or marker in the memory stack to detect buffer overflow attacks. These attacks try to overwrite a program's memory space, and a Canary acts as an early warning system.

Evaluating the effectiveness of Canaries involves assessing how well they detect and prevent stack buffer overflow vulnerabilities. Their limitations may include the possibility of bypassing or disabling them. There are situations where attackers might find ways to go around or turn off these warning signals then the defence mechanism becomes less reliable and may not effectively stop the threat.

2) Explain what is meant by ASLR, and then evaluate the effectiveness and limitations of ASLR as a defence versus attacks on format string vulnerabilities.

ASLR stands for Address Space Layout Randomization, a security feature that randomizes the memory addresses of system components, making it challenging for attackers to predict the location of vulnerable code.

Evaluating ASLR's effectiveness involves understanding how well it protects against format string vulnerabilities, a type of attack that exploits improper input handling. The limitations may include potential bypass techniques or scenarios using specific command where ASLR is less effective.

3) Explain what is meant by a Executable Space Protection (also known as NX-bits/DEP/W^X), and then evaluate the effectiveness and limitations of Executable Space Protection as a defence versus attacks on format string vulnerabilities.

Executable Space Protection, also known as NX-bits, DEP (Data Execution Prevention), or W^X (Write XOR Execute), prevents code execution from certain regions of memory. It aims to thwart attacks that involve injecting and executing malicious code.

Evaluating the effectiveness of Executable Space Protection involves analysing its ability to defend against attacks on format string vulnerabilities. Limitations may include situations where attackers find ways to execute code in non-executable memory regions.